

ALTOONA-BLAIR COUNTY AP, PA, USA

WMO: 725126

Lat: 40.296N

Lon: 78.320W

Elev: 451

StdP: 96.02

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14736

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-12.3	-20.8	0.6	-13.4	-18.4	0.8	-11.2	12.6	-1.1	11.6	-1.9	4.4	300	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	31.2	22.2	29.8	21.5	28.4	20.9	23.8	28.9	22.9	27.7	22.2	26.5	3.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.9	26.5	21.4	16.9	25.4	20.8	16.3	24.7	73.4	28.9	70.0	27.7	67.0	26.4	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.7	9.0	8.0	DB	-18.2	33.7	3.7	2.1	-20.9	35.2	-23.0	36.5	-25.1	37.6	-27.7	39.2
(5)				WB	-18.8	25.5	3.6	1.0	-21.3	26.2	-23.4	26.8	-25.4	27.4	-28.0	28.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-2.4	-1.0	3.5	10.2	15.6	20.1	22.1	21.2	17.6	11.6	5.6	0.5
	DBStd	9.82	6.39	5.79	5.86	5.16	4.60	3.38	2.87	2.72	3.75	4.69	5.04	5.25
	HDD10.0	1461	387	309	216	58	6	0	0	0	1	37	149	299
	HDD18.3	3240	642	540	459	247	111	22	4	7	59	214	382	553
	CDD10.0	1623	3	2	16	66	180	303	374	346	227	87	17	4
	CDD18.3	362	0	0	1	4	27	74	119	94	36	7	0	0
	CDH23.3	2784	0	0	7	47	230	548	949	687	276	39	1	0
	CDH26.7	740	0	0	1	6	51	138	290	179	69	5	0	0
Wind	WSAvg	3.2	4.1	4.1	4.0	3.9	3.1	2.8	2.3	2.0	2.2	3.0	3.5	3.7
Precipitation	PrecAvg	925	56	55	81	86	94	94	88	77	84	67	83	61
	PrecMax	1456	176	117	159	190	161	330	222	180	201	214	277	127
	PrecMin	617	7	16	13	13	11	14	25	0	8	2	14	10
	PrecStd	198	33	30	32	48	39	58	47	40	42	41	53	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	17.5	23.4	27.6	30.5	31.9	33.7	32.4	31.3	27.2	21.3	17.6
		MCWB	13.6	11.7	14.9	17.3	20.4	22.9	23.2	23.1	21.1	19.7	14.5	13.5
	2%	DB	13.4	13.2	19.5	24.5	28.0	29.9	31.5	30.3	28.8	24.0	17.9	13.9
		MCWB	10.7	8.9	12.4	15.3	19.1	21.7	22.5	22.0	20.5	17.8	12.6	11.0
	5%	DB	9.9	9.9	15.9	21.6	26.0	28.3	29.7	28.6	26.5	21.8	15.8	10.8
		MCWB	7.5	6.2	10.9	14.1	18.2	21.1	21.8	21.4	19.3	16.6	11.6	8.0
	10%	DB	6.3	7.1	12.8	18.9	23.7	26.6	28.0	27.1	24.4	19.4	13.6	7.7
		MCWB	4.2	4.2	8.6	12.5	17.3	20.2	21.2	20.7	18.7	15.3	10.2	5.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	13.1	16.3	19.0	21.9	24.5	25.0	24.7	23.4	21.1	16.1	14.4
		MCDB	16.0	16.3	22.3	24.6	28.0	29.1	31.2	29.9	28.2	24.9	18.9	17.1
	2%	WB	11.3	9.4	13.7	16.8	20.5	23.0	23.9	23.5	21.9	19.5	14.1	11.4
		MCDB	13.0	12.6	17.8	21.9	25.9	27.8	29.3	28.3	25.8	23.1	16.8	13.5
	5%	WB	7.5	6.6	11.5	15.2	19.5	22.1	23.0	22.6	20.9	17.5	12.4	8.3
		MCDB	9.5	9.3	15.2	20.1	24.5	26.6	27.9	26.8	24.4	20.1	14.9	10.1
	10%	WB	4.2	4.4	8.7	13.7	18.2	21.2	22.2	21.7	19.9	15.7	10.4	5.3
		MCDB	6.3	6.9	12.6	18.2	22.5	25.4	26.5	25.4	23.0	18.8	13.3	7.7
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.1	9.3	11.1	10.7	10.2	10.8	10.9	10.8	9.9	8.4	6.9
		MCDBR	10.1	12.1	13.7	14.5	13.5	12.1	12.6	12.6	13.7	13.3	11.9	10.3
		MCWBR	8.4	9.0	8.7	8.1	6.5	5.7	5.3	5.8	6.4	7.4	8.2	8.2
	5% WB	MCDBR	9.7	10.9	12.5	12.3	11.7	10.6	10.4	10.6	10.3	10.7	10.2	9.6
		MCWBR	8.7	8.7	8.8	8.1	6.1	5.5	5.0	5.5	5.5	6.7	8.3	8.5
Clear-Sky Solar Irradiance	taub	0.307	0.322	0.347	0.373	0.410	0.463	0.480	0.474	0.392	0.364	0.320	0.303	
	taud	2.425	2.353	2.376	2.340	2.250	2.128	2.096	2.097	2.311	2.391	2.495	2.473	
	Ebn at Noon	855	891	905	901	869	820	802	794	845	832	836	832	
	Edh at Noon	81	100	109	121	136	154	158	153	116	96	75	72	
All-Sky Solar Radiation	RadAvg	1.67	2.53	3.45	4.58	5.14	5.68	5.70	5.09	4.15	2.89	1.99	1.39	
	RadStd	0.13	0.31	0.28	0.42	0.52	0.47	0.37	0.29	0.51	0.33	0.22	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	+58

Nomenclature: See separate page